SALES PERFORMANCE AND PROFITABILITY ANALYSIS

Power BI portfolio project | Reporting period: 1 January–31 December 2025

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TRANSACTIONS	UNITS SOLD	TOTAL REVENUE	TOTAL PROFIT*	PROFIT MARGIN*
400	6,180	1,195,350	1,062,648	88.9%

Executive Summary

This report presents a sales and profitability analysis built in Power BI from the supplied sales, product, customer and supplier workbooks. The work included Power Query data preparation, data modelling and an interactive dashboard to monitor sales volume, revenue, product-level cost and calculated profit. Across 400 transactions in 2025, the dataset records 6,180 units sold, revenue of 1,195,350 and calculated product-level profit of 1,062,648. The resulting profit margin is 88.9%, calculated as total profit divided by total revenue. Monetary values are reported without a currency label because the supplied workbooks do not specify a currency.

*Profit and margin are calculated as sales revenue less product cost (quantity × unit cost). They should be interpreted as product-level profit and margin, not net profit after operating expenses, tax or other business costs.

Project Objectives

• Consolidate sales transactions with product, customer, date and supplier reference data.
• Prepare the data in Power Query and establish a usable Power BI data model.
• Create KPIs and visuals that compare revenue, profit, sales volume and performance across months, categories, sales representatives and suppliers.
• Provide transaction-level detail to support exploration and validation.

Data and Preparation

The source files comprise 400 sales records, 50 customer records, 10 product records and 6 supplier records. The sales table contains transaction ID, transaction date, product ID, customer ID, quantity, unit price, sales representative and promotion code. Product reference data supplies product name, category, unit cost, discontinuation status and supplier contact. Customer and supplier files provide descriptive attributes for slicing and analysis.

Power Query and Data Modelling

The Power BI workflow used Power Query to prepare the source tables and a relational model to connect transactions with the product, customer and date reference tables. Supplier analysis uses supplier contact fields to connect product-level supplier assignments to supplier names. The dashboard includes slicers for month, product category, sales representative and supplier. This report describes the completed workflow at a high level; individual transformation steps should be documented from the actual Power Query steps if a technical audit trail is required.

Key Measures

Measure	Definition	Result
Transactions	Count of sales records	400
Units sold	Sum of transaction quantity	6,180
Revenue	Sum of quantity × unit price	1,195,350

Measure	Definition	Result
Product cost	Sum of quantity × unit cost	132,702
Product-level profit	Revenue – product cost	1,062,648
Profit margin	Product-level profit ÷ revenue	88.9%
Loss-making transactions	Transactions where calculated profit < 0	44 (11.0%)

Performance Findings

1. Category performance

Widgets generated the highest revenue at 470,284 and the highest calculated profit at 423,294. Gadgets contributed 412,429 in revenue and 384,062 in profit. Tools generated 312,637 in revenue and 255,292 in profit. Widgets represented 39.3% of revenue, while Gadgets represented 34.5% and Tools 26.2%.

Category	Revenue	Product cost	Profit	Units
Widgets	470,284	46,990	423,294	2,547
Gadgets	412,429	28,367	384,062	1,913
Tools	312,637	57,345	255,292	1,720

2. Sales representative performance

Rep_5 recorded the highest revenue at 169,767, accounting for 14.2% of total revenue. The dashboard's sales-representative bar chart supports comparison across all representatives, while the slicers allow performance to be examined by month, category and supplier.

Sales representative	Revenue	Profit	Units
Rep_5	169,767	153,587	749
Rep_8	147,934	129,824	748
Rep_10	138,392	120,666	735
Rep_2	119,267	107,392	489
Rep_6	116,534	101,688	691

3. Monthly trend

January recorded the highest monthly revenue at 160,176 and the highest monthly profit at 140,184. September recorded the lowest monthly revenue at 50,256. The monthly trend visual makes changes in revenue and product-level profit visible across the year; the chart should use chronologically sorted month labels.

Month	Revenue	Profit	Units
Jan 2025	160,176	140,184	763
Feb 2025	127,100	114,971	644
Mar 2025	70,608	62,734	407
Apr 2025	114,342	106,852	444
May 2025	90,227	76,062	600
Jun 2025	79,392	71,507	417
Jul 2025	127,859	115,734	543
Aug 2025	116,937	105,768	540
Sep 2025	50,256	44,127	308
Oct 2025	94,921	81,012	563
Nov 2025	105,156	93,782	509

Month	Revenue	Profit	Units
Dec 2025	58,376	49,915	442

Supplier and Product Findings

4. Supplier contribution

Supplier contribution is evaluated using product supplier contacts matched to the supplier reference data. WhatsUp3 has the highest aggregated supplier profit at 523,335. The source supplier file contains multiple records labelled WhatsUp3 associated with different old supplier contacts; the dashboard aggregates those records under the same supplier name. This aggregation should be understood when interpreting supplier-level totals.

Supplier	Revenue	Product cost	Profit	Units
WhatsUp3	573,274	49,939	523,335	2,758
The MEGA Coup	298,482	43,243	255,239	1,622
ESCANOR55	201,368	32,040	169,328	1,120
HeyBo4	122,226	7,480	114,746	680

5. Product-level observations

Product 3 (P003) is the highest-revenue individual product in the supplied transactions, with revenue of 164,845. The product analysis can be extended to compare sales volume, revenue and product-level profit by product ID and category. Product discontinuation flags are present in the product source data but are not, by themselves, evidence that a discontinued product should be excluded from historical sales analysis.

Product	Category	Revenue	Product cost	Profit	Units
Product 3 (P003)	Gadgets	164,845	13,946	150,899	734
Product 8 (P008)	Widgets	146,690	7,470	139,220	747
Product 5 (P005)	Tools	134,428	23,868	110,560	612
Product 7 (P007)	Gadgets	127,311	4,655	122,656	665
Product 2 (P002)	Widgets	124,437	16,500	107,937	750

Dashboard Design and Interactivity

The dashboard combines four headline KPIs—Total Revenue, Total Profit, Total Units Sold and Profit Margin—with slicers for Month, Product Category, Sales Representative and Supplier. Supporting visuals show revenue by sales representative, monthly revenue and profit trend, profit by product category, profit contribution by supplier and revenue contribution by category. A transaction table provides row-level detail for investigation. The visual design uses differentiated colour panels to separate sections while keeping the KPIs prominent.

Recommendations

• Monitor monthly revenue and product-level profit together to identify periods where sales and profitability move differently.
• Review the category mix: Widgets currently leads the dataset in both revenue and calculated profit; track whether this pattern persists in later periods.
• Compare sales-representative performance alongside volume and profit, rather than using revenue alone.
• Review supplier mappings and the repeated WhatsUp3 supplier name to ensure the intended aggregation is documented.
• Add a loss-making transaction table only after transaction-level profit is validated against quantity, unit price and product unit cost.
• For publication, remove the dollar symbol unless the source currency is confirmed, and replace abbreviated KPI labels with exact values where space allows.

Limitations and Validation Notes

The analysis covers the 2025 transaction dates present in the supplied sales file. Product cost is calculated as quantity multiplied by the unit cost in the product reference table. The reported profit margin of 88.9% is therefore a product-level margin and excludes operating expenses, tax, freight, discounts not represented in the supplied fields and other costs not included in the workbooks. The source files do not define a currency. The source contains 0 unmatched product records and 0 unmatched supplier mappings after the joins used for this analysis; all supplied sales rows matched to a product and supplier contact in the calculated dataset. Any final publication should verify the Power BI relationships, aggregation settings and supplier-name grouping against the report model.

Conclusion

The Power BI dashboard provides an interactive view of sales performance and product-level profitability across the 2025 dataset. It consolidates multiple reference files, supports filtering across key business dimensions and makes it easier to compare monthly trends, category contribution, sales-representative performance and supplier contribution. The main next step is validation of transaction-level calculations and final formatting before sharing the dashboard as a portfolio project.